

Shaoming Wu

Phone: 647-327-1398 — Email: shaoming.wu@outlook.com — LinkedIn: [shaoming-wu](#) — GitHub: [shaoming11](#)

Education

Milliken Mills High School

IB Programme

September 2022 – June 2026

Markham, ON

University Of Waterloo

Mechatronics Engineering

September 2026 – June 2031

Waterloo, ON

Experience

Co-Founder and Technical Lead

Altura Robotics (VEX Robotics Team 1165A)

May 2024 – Present

Markham, ON

- Led a team of 12 developers using GitHub workflows to design and deliver competition-ready software systems
- Improved **autonomous performance by 40%** by developing and tuning **C++ control algorithms** and iterative testing processes
- **Reduced debugging time by 50%** by establishing coding standards, documentation, and structured code review practices
- Organized a **40-team robotics tournament** by coordinating 20+ volunteers, delivering a 100+ participant event on schedule

Outreach Officer

Hack Canada

September 2021 – May 2024

Waterloo, ON

- Collaborated with a 30-member organizing team to plan and execute a 400-person technology hackathon
- Secured sponsorships through partner outreach and relationship management to support event operations
- Supported logistics and participant experience to deliver a large-scale innovation event for students across Ontario

Projects

Smooth Particle Hydrodynamics Simulator | *Python, NumPy*

Mar 2025

- Collaborated with PhD students at the University of Toronto to investigate **smooth-particle-hydrodynamics** on Google Colab and Jupyter Notebook
- Simulated **100+ particles for 205% increased processing speed**

helpidontknowhowtonetworkin.tech | *MERN Stack, TensorFlow*

May 2025

- Developed fullstack face recognition app with 95% accuracy by using React, Node.js, Express.js, and MongoDB for 60% faster NoSQL queries
- Recipient of the **Best Use of Generative AI at JAMHacks 9**

SupplyMe | *Django REST API, Flutter, MongoDB*

May 2025

- Constructed a vector embedding search database tool with **90% search accuracy**
- Developed front-end interface with Flutter, leading to load times of **< 5ms**

shao-lib | *C++*

Jan 2026

- Minimized error by 10x in a robotics drivetrain by developing an **Unscented Kalman Filter** for state estimation in VEX Robotics.
- Used 4 LiDAR sensors, 2 IMUs and a rotation sensor for position tracking. Integrated **sensor fusion for accuracy-based weightings**, improving accuracy by 210%.

Technical Skills

Languages: JavaScript, React, Python, C/C++, HTML/CSS, TypeScript

Frameworks & Libraries: Express.js, Node.js, Django, Flutter, TensorFlow, PostgreSQL

Databases: MongoDB, Supabase

CAD Software: Onshape, Inventor, Fusion 360